

ML 101

Regression, classification & core machine learning concepts

Name _____

Date _____

Score ____ / 15

Instructions Choose one answer for each multiple-choice question. Questions 13-15 are short answer. Suggested time: 20-25 minutes. Each question is worth 1 point.

1. Which task is a regression problem?

- A. Predict whether an email is spam
- B. Predict the price of a used car
- C. Identify which animal appears in a photo
- D. Decide whether a customer will repay a loan

2. Which task is a classification problem?

- A. Estimate tomorrow's temperature
- B. Predict the time a delivery will take
- C. Predict whether a transaction is fraudulent
- D. Estimate a house's selling price

3. In supervised learning, what does a training example usually contain?

- A. Features only
- B. A feature set and a known target label/value
- C. A model and its final prediction only
- D. Unlabelled data and no target

4. What is the main purpose of splitting data into training and test sets?

- A. Make the dataset larger
- B. Measure how well the model works on unseen data
- C. Ensure the training score is high
- D. Remove all noisy observations

5. A model gets very low error on training data but much higher error on test data. What is the most likely issue?

- A. Underfitting
- B. Overfitting
- C. Data normalization
- D. Classification

6. Which change is most likely to reduce overfitting?

- A. Add more complexity to the model
- B. Evaluate only on training data
- C. Use regularization or a simpler model
- D. Remove the test set

7. What is a loss function used for?

- A. To quantify how wrong a model's predictions are
- B. To split data into train and test sets
- C. To choose which rows belong in the dataset
- D. To guarantee a model is fair

8. For predicting a numeric value, which loss is commonly used?

- A. Mean squared error
- B. Accuracy
- C. Recall
- D. A confusion matrix

9. Gradient descent is a method for...

- A. Choosing random labels for training
- B. Adjusting model parameters to reduce loss
- C. Separating training and test data
- D. Turning regression into classification

10. During gradient descent, what does the learning rate control?

- A. The size of each parameter update
- B. The number of classes
- C. The fraction of data used for testing
- D. The number of input features

11. A model predicts “positive” for 10 cases; 7 are truly positive. What is its precision?

- A. $7/10 = 70\%$
- B. 7 divided by all actual positives
- C. $10/7$
- D. 30%

12. Why should you avoid repeatedly tuning a model based on its test-set score?

- A. Test sets contain no useful information
- B. You can indirectly overfit to the test set, making its score less independent
- C. It changes the target values
- D. It makes gradient descent impossible

Short answer

Answer in a few sentences or with a small sketch.

13. Explain the difference between a model's training set and test set. Why do we keep the test set separate?

14. A model is too simple to capture the pattern in both its training and test data, and both errors are high. Is this overfitting or underfitting? Give one way to improve it.

15. In plain language, describe how gradient descent uses the loss to improve a model. What might happen if the learning rate is far too large?
